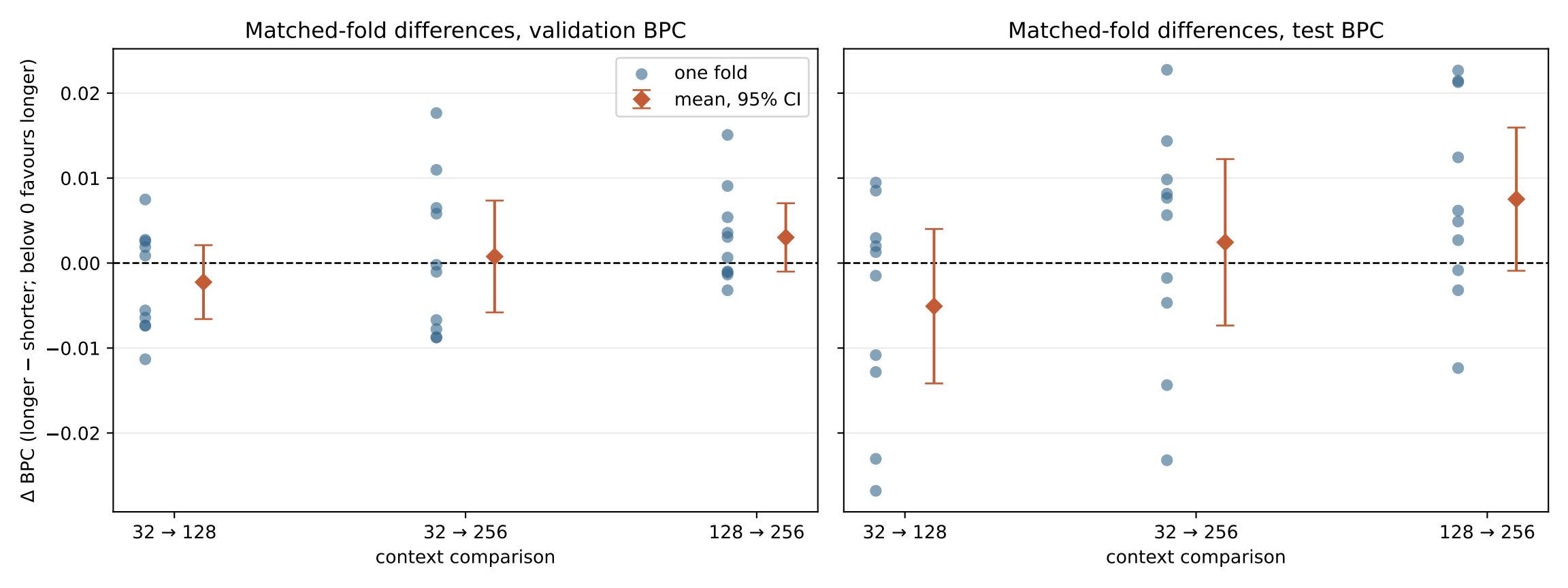
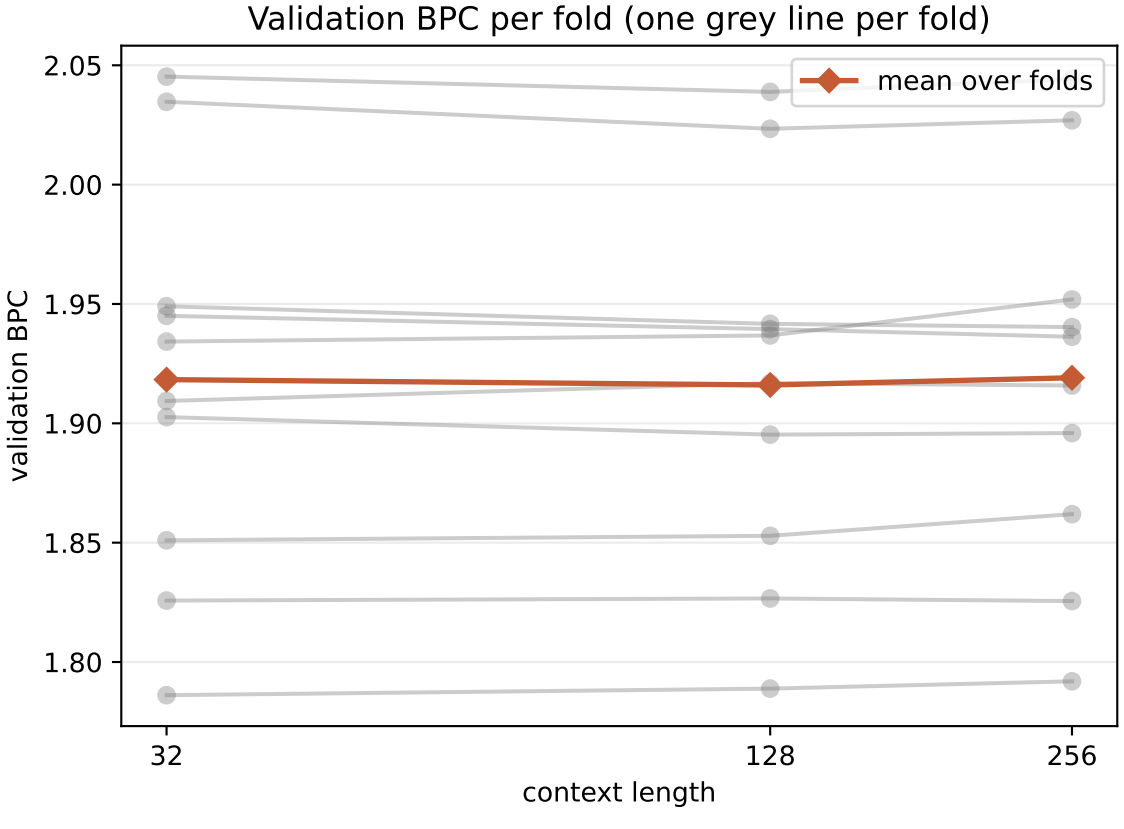
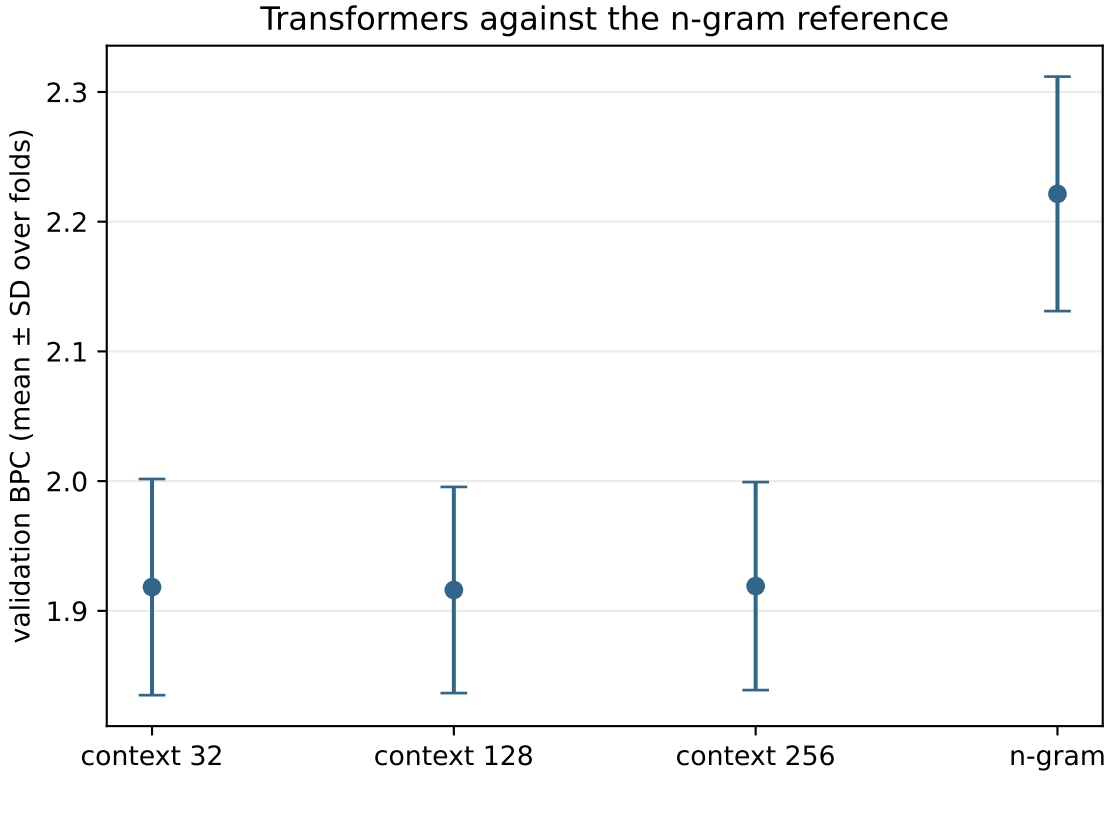
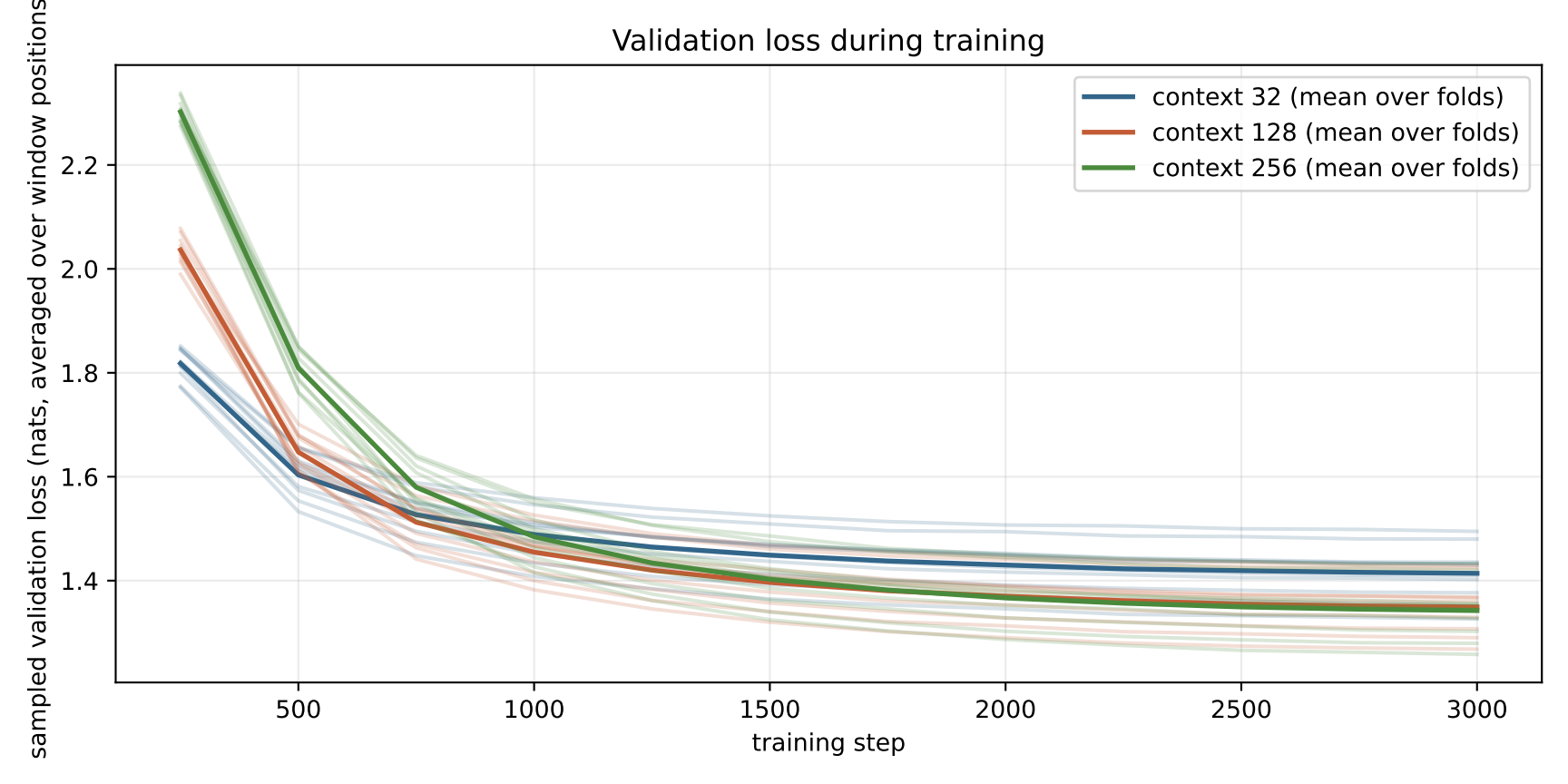
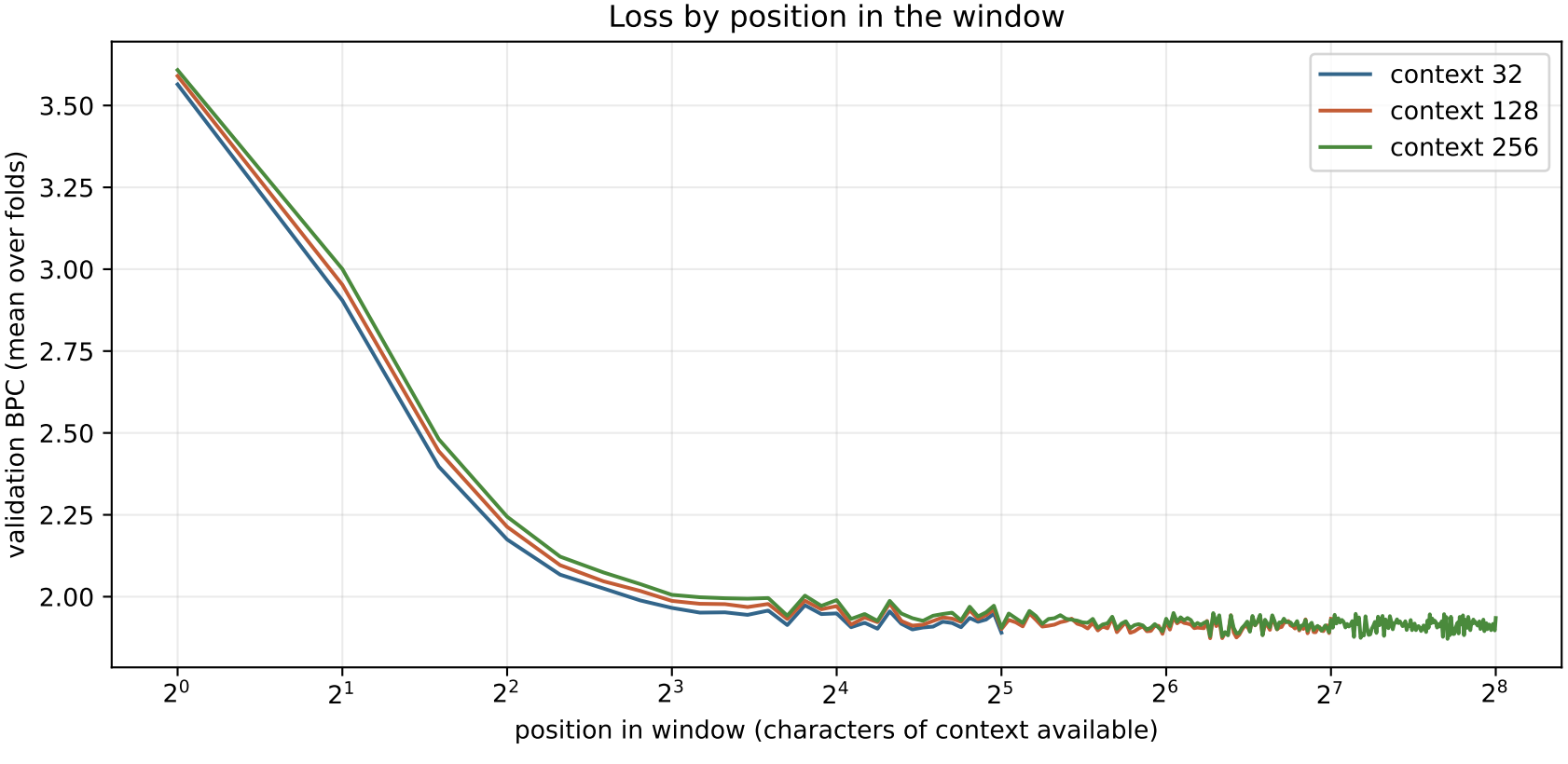
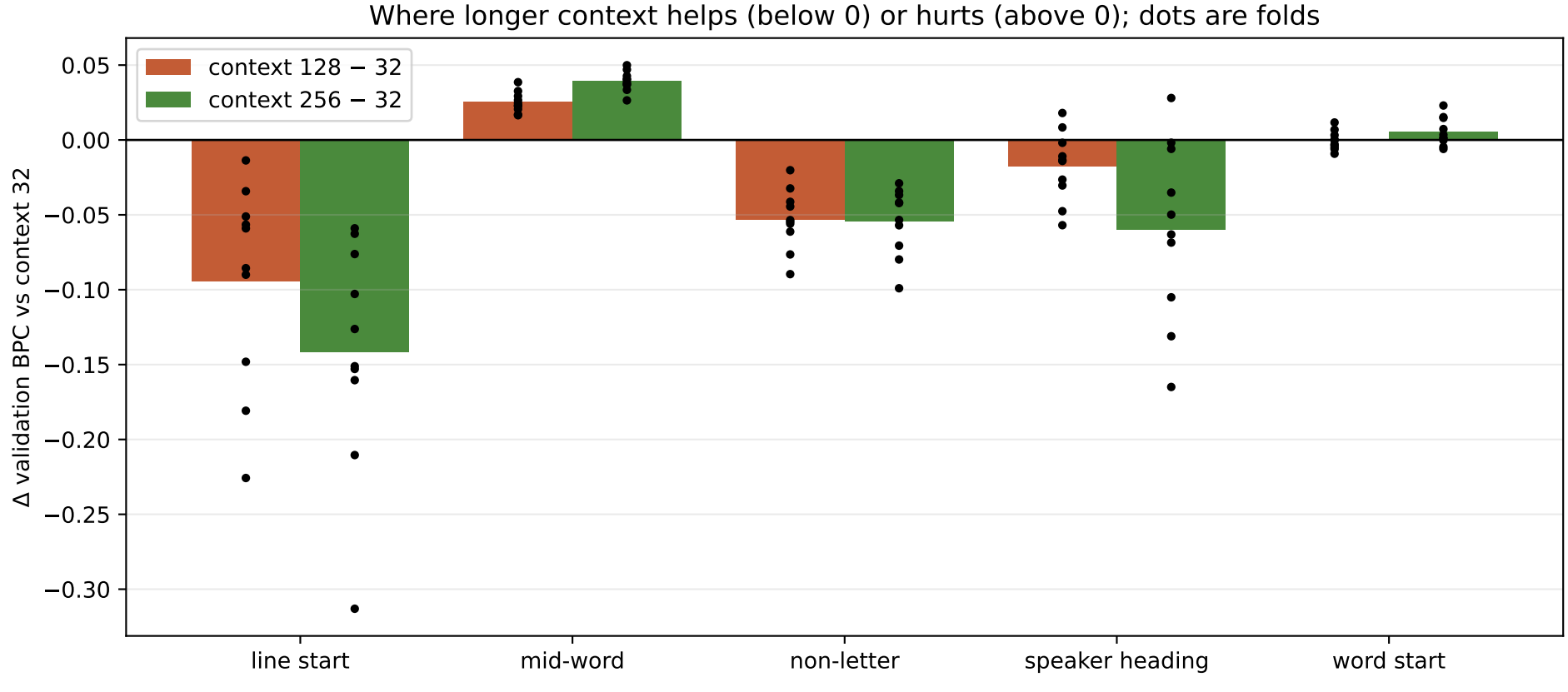

Matched-fold differences, validation BPC
Matched-fold differences, test BPC
one fold
mean, 95% CI
Δ BPC (longer − shorter; below 0 favours longer)
0.02
0.01
0.00
−0.01
−0.02
32 → 128
32 → 256
128 → 256
context comparison


Validation BPC per fold (one grey line per fold)
mean over folds
validation BPC
2.05
2.00
1.95
1.90
1.85
1.80
32
128
256
context length


Transformers against the n-gram reference
validation BPC (mean ± SD over folds)
2.3
2.2
2.1
2.0
1.9
context 32
context 128
context 256
n-gram


Validation loss during training
context 32 (mean over folds)
context 128 (mean over folds)
context 256 (mean over folds)
sampled validation loss (nats, averaged over window positions)
2.2
2.0
1.8
1.6
1.4
500
1000
1500
2000
2500
3000
training step


Loss by position in the window
context 32
context 128
context 256
validation BPC (mean over folds)
3.50
3.25
3.00
2.75
2.50
2.25
2.00
2^0
2^1
2^2
2^3
2^4
2^5
2^6
2^7
2^8
position in window (characters of context available)


Where longer context helps (below 0) or hurts (above 0); dots are folds
context 128 − 32
context 256 − 32
Δ validation BPC vs context 32
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
line start
mid-word
non-letter
speaker heading
word start